Machine Learning for Residential Property Value Prediction in the French Real Estate Market: Geographic, Energy, and Temporal Effects

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***Abstract*: Accurate residential property valuation is challenging because real estate prices depend simultaneously on physical characteristics, location, market conditions, and other contextual factors. This study investigates residential sale-price prediction in France using open data and machine learning. A national dataset of 708,520 exploitable residential transactions from 2024 is constructed from DVF+ after restricting the data to sales containing exactly one dwelling, excluding mixed residential–commercial transactions, and removing observations with unusable physical characteristics or target values.**

A naive median baseline, Linear Regression, Random Forest, and Gradient Boosting are compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Four hypotheses are then evaluated: whether Gradient Boosting outperforms simpler approaches; whether geographic information improves accuracy; whether the French energy-performance label (DPE) provides additional predictive information; and whether temporal generalization is more difficult than a random within-year split.

The best national configuration is Gradient Boosting with department-level geographic information, reaching a MAE of EUR 95,345, an RMSE of EUR 222,285, and an R^2 of 0.421. Geography produces a substantial improvement over the same model without department information. A separate DPE experiment is conducted on 22,641 Parisian transactions matched with ADEME certificates through geographic proximity, temporal consistency, and dwelling-surface similarity; adding the DPE label does not improve prediction accuracy on this subset. Finally, training on 2023 transactions and evaluating on 2024 causes only a moderate performance degradation. Feature-importance analysis shows that living area and department-level geography dominate the national Gradient Boosting model.

***Index Terms*: real estate, property valuation, machine learning, Gradient Boosting, DVF+, DPE, geographic information, temporal generalization, France**

I. Introduction

Automated valuation models have become increasingly relevant to real estate professionals, financial institutions, public authorities, and households. Property prices are affected by multiple interacting factors, including dwelling size, property type, land characteristics, local market conditions, and geographic location. These relationships are often nonlinear, making residential valuation a natural application for machine-learning regression.

The French market is particularly suitable for reproducible research because several nationwide public datasets are available. The DVF+ open-data product released by Cerema restructures the French property-transaction database into transaction-level records and provides geographic representations of mutations [4]. In parallel, ADEME publishes the national database of Diagnostic de Performance Energétique (DPE) certificates for existing dwellings established since July 2021 [5]. These sources make it possible to investigate not only predictive performance, but also the incremental value of geographic and energy information.

Previous work has shown strong performance of tree-based ensemble methods on structured real-estate data. Sharma, Harsora, and Ogunleye compared several regression algorithms on the Ames housing dataset and reported XGBoost as the best-performing approach [1]. Neves, Aparicio, and de Castro Neto demonstrated that contextual open data can improve real-estate price prediction in Lisbon and used explainable-AI methods to interpret the resulting model [2]. Chen *et al.* studied rental-value prediction in Munich and similarly found ensemble methods to be competitive for tabular real-estate data [3]. These studies motivate the use of boosting methods while also highlighting the importance of geographic and contextual information.

However, three issues remain particularly relevant for a French nationwide setting. First, model comparisons performed on a single city or highly detailed private dataset do not directly indicate how well simple open-data features perform across heterogeneous French territories. Second, the additional predictive value of DPE information should be evaluated on the same transactions rather than inferred from descriptive correlations. Third, random train/test splits may overestimate real-world performance when a model is ultimately used to predict transactions from a later period.

The main research question of this study is therefore:

> *Which machine-learning approach provides the most accurate prediction of residential property sale prices in France using open data?*

Four hypotheses are evaluated:

- **H1:** Gradient Boosting outperforms simpler regression approaches for residential property price prediction.
- **H2:** Geographic information substantially improves prediction accuracy compared with models based only on physical property characteristics.
- **H3:** Adding DPE energy-performance information improves prediction accuracy beyond physical and geographic variables.
- **H4:** Temporal generalization, where a model trained on past transactions predicts a subsequent year, is more difficult than a random train/test split within the same year.

The contributions of this study are: (i) the construction of a nationwide 2024 modeling dataset containing 708,520 exploitable residential transactions; (ii) a controlled comparison of a naive baseline, Linear Regression, Random Forest, and Gradient Boosting; (iii) an incremental evaluation of department-level geography; (iv) a reproducible DVF+–DPE matching procedure combining geographic proximity, temporal consistency, and surface similarity; (v) an empirical DPE ablation on 22,641 matched Parisian transactions; (vi) a 2023-to-2024 temporal generalization experiment; and (vii) an interpretation of the best national model through aggregated feature importance.

II. Related Work

A. Machine Learning for Property Valuation

Real-estate valuation has traditionally relied on hedonic regression and comparable-sales methods. Machine-learning methods can model nonlinearities and interactions without requiring the functional form to be specified in advance. Sharma *et al.* [1] compared XGBoost, Support Vector Regression, Random Forest, Multilayer Perceptron, and Multiple Linear Regression on the Ames housing dataset. Their results favored XGBoost, reinforcing the suitability of boosting methods for structured housing data. Their setting, however, concerns a single U.S. city with a rich set of property attributes, whereas the present study addresses nationwide French transactions with a deliberately compact open-data feature set.

B. Contextual Open Data and Geography

Neves *et al.* [2] studied Lisbon transactions from 2018 to 2021 and enriched an XGBoost model with contextual open data. Their results showed that transport, points of interest, socio-demographic information, and other spatial features can improve prediction accuracy. The study also used explainable AI to examine feature contributions. This supports the hypothesis that location should be explicitly represented, but the present work tests this proposition first at the coarser department level over the whole of France.

C. Energy and Rental-Oriented Research

Chen *et al.* [3] investigated rental-price prediction in Munich using several machine-learning techniques and reported strong performance from ensemble models. Although the prediction target differs from the present study, their findings provide additional evidence that tree ensembles are well suited to real-estate tabular data. In France, DPE certificates constitute an important regulated source of dwelling-level energy information. ADEME notes that the DPE database contains raw diagnostics and does not represent the complete housing stock, so interpretation requires caution [5]. This motivates treating DPE as an incremental predictive feature on a controlled matched subset rather than assuming universal coverage.

D. Positioning of the Present Study

The present work differs from prior studies in four ways. It evaluates a nationwide French sample rather than a single local market; it performs controlled incremental tests of geography and energy information; it explicitly evaluates future-year generalization; and it documents a practical procedure for matching transaction-level DVF+ records with DPE certificates when no direct common dwelling identifier is available.

III. Methodology

A. Data Sources

DVF+. The principal source is DVF+ open data produced by Cerema from the French DVF property-transaction database. Cerema provides transaction-level CSV files and GeoPackage files in which each entity corresponds to a sale associated with a geometry [4]. Departmental CSV files from the 2014–2024 delivery are used to construct the national 2024 and 2023 samples.

DPE. Energy-performance information comes from ADEME's "DPE Logements existants (depuis juillet 2021)" dataset. It contains the DPE label, greenhouse-gas label, dwelling surface, certificate date, geocoded address information, and geographic coordinates, among many other fields [5]. The DPE data are accessed through the public API.

B. Construction of the National 2024 Dataset

The 97 departmental DVF+ mutation files are combined after selecting transactions with `anneemut = 2024` and `libnatmut = Vente`. This produces 1,129,669 sale records.

The objective is to associate each target price with one principal residential dwelling. For each sale, the number of dwellings is reconstructed as

$$N_{\text{dwelling}} = nblocmai + nblocapt, \tag{1}$$

where `nblocmai` and `nblocapt` are the numbers of houses and apartments included in the mutation. Only sales with exactly one dwelling are retained, leaving 723,545 transactions. Mixed transactions containing commercial or activity premises are then removed by imposing `nblocact = 0`, leaving 709,820 residential-only sales.

The dwelling surface is defined as the sum of house and apartment built surfaces:

$$S_{\text{dwelling}} = sbatmai + sbatapt. \tag{2}$$

The number of principal rooms is reconstructed from the DVF+ room-count fields, and property type is defined as

House when the transaction contains one house and *Apartment* when it contains one apartment. Transactions with non-positive dwelling surface or room count are removed. Finally, records with missing transaction value or a value not exceeding EUR 1,000 are excluded. The resulting national 2024 modeling dataset contains 708,520 sales.

The retained variables are:

- transaction value (`valeurfonc`), used as the target;
- dwelling surface;
- number of principal rooms;
- property type;
- number of dependencies (`nblocdep`);
- land surface (`sterr`);
- department code (`coddep`);
- municipality/INSEE information for selected experiments;
- mutation date (`datemut`) for temporal evaluation.

Categorical variables are one-hot encoded. No target-derived feature such as local price per square meter is used, thereby avoiding direct leakage from the dependent variable.

C. Experimental Protocol

For the main 2024 experiments, the data are split randomly into 80% training and 20% testing sets with `random_state=42`, resulting in 566,816 training transactions and 141,704 test transactions.

Three evaluation metrics are used:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{3}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{4}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{5}$$

MAE is the primary comparison metric because it is directly interpretable in euros and is less dominated by extreme errors than RMSE. RMSE and R^2 are reported as complementary measures.

D. Models and H1

Four approaches are compared on the same physical feature set:

- a **naive baseline** predicting the median price of the training set for every test transaction;
- **Linear Regression**;
- **Random Forest Regression**;
- **Gradient Boosting Regression**.

The physical feature set contains dwelling surface, room count, property type, number of dependencies, and land surface. The Gradient Boosting experiments use 200 estimators, a learning rate of 0.05, maximum tree depth of 3, and `random_state=42`. Hyperparameter optimization is intentionally limited; the goal is a controlled comparison and incremental feature study rather than exhaustive leaderboard optimization.

E. Geographic Increment and H2

To test H2, department code is added to the Gradient Boosting feature set and one-hot encoded. The resulting matrix contains 101 explanatory columns. The same 2024 split and evaluation metrics are used, allowing direct comparison between Gradient Boosting with physical characteristics only and Gradient Boosting with physical characteristics plus department-level geography.

F. DPE Matching and H3

Because DVF+ and ADEME DPE do not provide a direct common dwelling identifier, H3 requires a controlled matching procedure. To keep the experiment computationally tractable and geographically coherent, the DPE analysis is performed on Paris.

The final national dataset contains 25,357 Parisian transactions. The April 2025 DVF+ Paris GeoPackage is used to recover transaction geometry from the `mutation_geomlocmut` layer through `idmutation`. A total of 25,356 sales are successfully geolocated. DVF+ geometries are originally expressed in Lambert-93 (EPSG:2154) and are converted when necessary for interoperability.

The ADEME API is used to retrieve DPE records for the 20 Paris arrondissements. A total of 371,600 certificates are collected. Certificates established after 2024 are initially removed, leaving 291,345 records; records without dwelling surface are then excluded, leaving 287,650 DPE records with usable surface and geographic coordinates.

Candidate matches are generated within a 50 m radius around each sale. A DPE is eligible only if its establishment date is on or before the exact mutation date, preventing the use of future information. This produces at least one temporally valid nearby DPE for 24,965 sales (98.46% of geolocated Parisian sales). Candidate quality is then assessed using the relative surface difference:

$$\Delta_S(\%) = \frac{|S_{\mathrm{DVF+}} - S_{\mathrm{DPE}}|}{S_{\mathrm{DVF+}}} \times 100. \tag{6}$$

A maximum surface difference of 10% is retained. When several candidates remain, they are ranked by smaller surface difference, then smaller geographic distance, then more recent valid DPE date. The final matched H3 dataset contains 22,641 transactions, corresponding to 89.29% of the geolocated Parisian sample.

To isolate the incremental effect of the DPE label, the 22,641 matched transactions are split once into 18,112 training and 4,529 test observations. The same rows and Gradient Boosting parameters are used in both configurations. The baseline H3 model uses physical variables and Paris arrondissement; the second configuration adds the one-hot encoded DPE label.

G. Temporal Generalization and H4

The 2023 national dataset is reconstructed using exactly the same filtering and feature-engineering rules as 2024.

It contains 784,582 exploitable residential transactions. The temporal experiment trains Gradient Boosting on the entire 2023 dataset and evaluates it on the entire 2024 dataset. After category alignment, both matrices contain the same 101 explanatory variables. This design approximates a realistic future-year prediction scenario.

H. Feature Importance

The best national model is interpreted using the built-in tree feature importances of Gradient Boosting. Because one-hot encoding expands department into many binary variables, individual department importances are aggregated into a single "departmental geography" family. The reported values describe how strongly variables are used by the fitted model; they are not causal effects.

I. Overall Experimental Design

IV. Results

A. H1: National Model Comparison

Table I reports the 2024 national results using physical property characteristics only.

TABLE I
NATIONAL 2024 MODEL COMPARISON WITHOUT DEPARTMENT GEOGRAPHY.

Model	MAE (EUR)	RMSE (EUR)	R^2
Gradient Boosting	119,235	279,881	0.0826
Linear Regression	124,345	266,822	0.1662
Random Forest	128,168	288,355	0.0262
Naive baseline	128,936	297,877	-0.0392

Gradient Boosting achieves the lowest MAE, improving the average absolute error by approximately EUR 9,700 relative to the naive baseline. However, Linear Regression achieves a better RMSE and a higher R^2 than Gradient Boosting. H1 is therefore only *partially supported*: Gradient Boosting is the best model according to the primary MAE criterion, but it does not dominate all evaluation metrics.

B. H2: Contribution of Geographic Information

Adding department geography to Gradient Boosting produces a large improvement, as shown in Table II.

TABLE II
IMPACT OF DEPARTMENT-LEVEL GEOGRAPHIC INFORMATION.

Configuration	MAE (EUR)	RMSE (EUR)	R^2
GB, physical only	119,235	279,881	0.0826
GB + department	**95,345**	**222,285**	**0.4213**

The MAE decreases by approximately EUR 23,890, corresponding to a reduction of about 20%. At the same time, R^2 rises from 0.083 to 0.421. H2 is therefore strongly supported: even a coarse geographic representation at department level contributes substantially to nationwide valuation accuracy.

C. H3: Contribution of DPE Information

The DPE experiment uses exactly the same 22,641 matched Parisian transactions in both configurations. Results are shown in Table III.

TABLE III
IMPACT OF THE DPE LABEL ON THE MATCHED PARISIAN SAMPLE.

Configuration	MAE (EUR)	RMSE (EUR)	R^2
Without DPE	**122,204**	**270,463**	**0.7697**
With DPE	122,401	272,650	0.7659

Adding the DPE label does not improve predictive performance. MAE increases slightly by approximately EUR 197, RMSE also increases, and R^2 decreases from 0.770 to 0.766. H3 is therefore not supported in this experiment. This finding does not imply that energy efficiency has no relationship with property value; rather, the categorical DPE label does not provide measurable incremental predictive value beyond the physical and geographic variables used on this matched Parisian subset.

D. H4: Temporal Generalization

Table IV compares the best 2024 random-split configuration with a model trained on 2023 and evaluated on 2024.

TABLE IV
RANDOM WITHIN-YEAR EVALUATION VERSUS TEMPORAL GENERALIZATION.

Configuration	MAE (EUR)	RMSE (EUR)	R^2
Random split 2024	**95,345**	**222,285**	**0.4213**
Train 2023 $\rightarrow$ Test 2024	96,551	238,338	0.4093

Temporal generalization is slightly worse. The MAE increases by approximately 1.3%, while RMSE increases by about 7.2%, indicating a larger increase in severe errors than in typical absolute errors. The R^2 decreases moderately from 0.421 to 0.409. H4 is therefore moderately supported.

E. Feature Importance of the Best National Model

Table V aggregates the Gradient Boosting feature importances into interpretable families.

TABLE V
AGGREGATED FEATURE IMPORTANCE OF THE BEST NATIONAL MODEL.

Feature family	Importance (%)
Dwelling surface	48.71
Departmental geography	44.86
Land surface	3.76
Dependencies	2.25
Number of rooms	0.41
Property type	0.01

Dwelling surface and departmental geography together account for approximately 93.6% of the model's aggregated

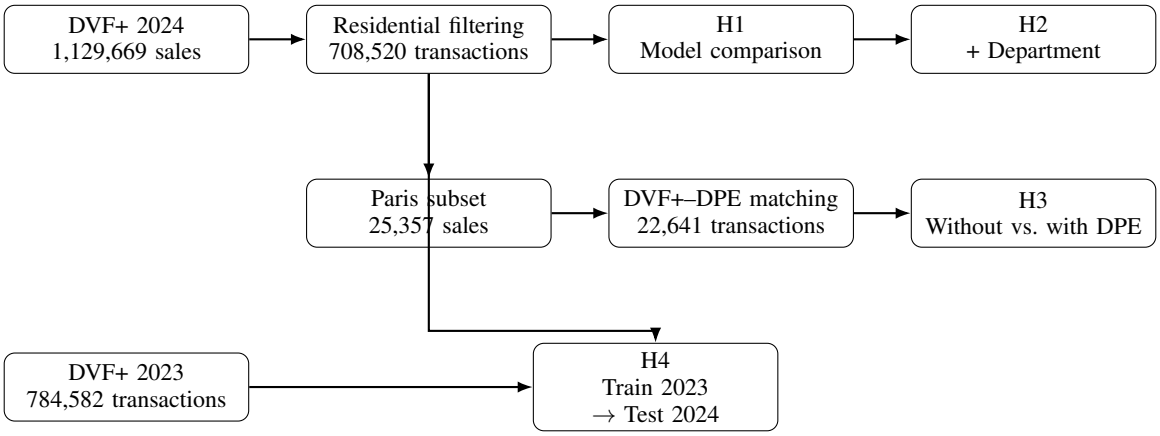


Fig. 1. Experimental workflow used to evaluate the four hypotheses.

importance. This result is consistent with H2: much of the predictive signal available in the selected open-data feature set is concentrated in property size and geographic location. The low marginal importance of room count and property type should be interpreted conditionally: once surface, geography, land area, and dependencies are already known, these variables add little additional information to this fitted model.

F. Summary of Hypotheses

TABLE VI
SUMMARY OF THE FOUR HYPOTHESES.

Hypothesis	Conclusion
H1	Partially supported
H2	Supported
H3	Not supported
H4	Moderately supported

V. DISCUSSION

A. Model Complexity and H1

The H1 results illustrate why model comparison should not be reduced to a single statement that one algorithm is universally superior. Gradient Boosting achieves the lowest MAE, which is the main operational metric of this study, but Linear Regression obtains a better RMSE and R^2. One explanation is that, without explicit geographic information, a flexible nonlinear model has limited contextual information from which to learn. The national price distribution is highly heterogeneous, and severe errors remain difficult to control with the compact physical feature set.

The result is nevertheless consistent with prior studies showing strong performance of boosting methods on structured housing data [1]. The key difference is that the present nationwide dataset contains much less detailed dwelling information than datasets such as Ames. Model sophistication cannot fully compensate for missing explanatory context.

B. Geography as the Main Incremental Signal

H2 is the clearest result of the study. Department information reduces MAE from approximately EUR 119,000 to EUR 95,000 and increases R^2 from 0.083 to 0.421. The feature-importance analysis leads to the same conclusion: department-level geography accounts for approximately 44.9% of the aggregated importance of the best model.

This finding is aligned with the contextual-data results of Neves *et al.* [2]. Their Lisbon study used considerably finer spatial information than the department representation used here. Consequently, the present results suggest that additional local context, municipality, neighborhood, accessibility, or spatial coordinates—is a promising direction for improving nationwide French valuation.

C. Why the DPE Label Did Not Improve H3

The H3 result is deliberately interpreted conservatively. On the matched Parisian sample, adding the categorical DPE label slightly worsens all three metrics. Several mechanisms may explain the lack of incremental predictive gain.

First, the DPE label may be partly correlated with characteristics already represented by dwelling surface, property type, and location. Second, Parisian prices are strongly driven by very fine spatial effects, so a categorical energy label may be small relative to local price heterogeneity. Third, the experiment uses only the DPE class, not the full energy-consumption and building-characteristic information available in the ADEME database. Finally, although the matching rules are strict, the absence of a shared dwelling identifier means some residual matching error cannot be excluded.

The conclusion is therefore specific: *the DPE label did not improve the tested predictive model on the matched Parisian sample*. It should not be generalized into a claim that energy performance has no economic effect on property value.

D. Temporal Stability

H4 shows that the model remains relatively stable when moving from 2023 to 2024. The MAE rises only modestly, but the larger increase in RMSE indicates that future-year

prediction produces more substantial errors for a subset of transactions. This distinction is important because a random split can place transactions from the same market regime in both training and test data, whereas temporal evaluation better approximates deployment.

The moderate degradation is encouraging for short-horizon generalization, but a single one-year transition cannot establish long-term stability. Evaluating several rolling train/test periods would provide stronger evidence.

E. Practical Interpretation

The best national model reaches an average absolute error of approximately EUR 95,000. This is a substantial improvement over the naive baseline, yet it remains too large to interpret the system as a replacement for professional appraisal. The model should instead be viewed as a large-scale statistical valuation baseline showing how much information can be extracted from a compact set of French open-data variables. Its strongest use in this research is comparative: quantifying the contribution of geography, energy information, and temporal evaluation under a controlled protocol.

VI. Limitations

Several limitations should be considered.

Geographic granularity. The national model represents location mainly through department code. Departments contain substantial internal price heterogeneity, especially in dense metropolitan areas. Finer geographic information is likely to improve accuracy.

DPE scope. H3 is evaluated only on Parisian transactions with a reliable DPE match. The result cannot automatically be generalized to other French territories, where the relationship between energy efficiency and price may differ.

Probabilistic DVF+–DPE matching. There is no shared dwelling identifier between the two open datasets. The matching procedure combines a 50 m geographic radius, a DPE date not later than the sale date, and a maximum 10% surface difference. These rules substantially constrain candidate matches but cannot guarantee perfect identity in every case.

Temporal coverage. The principal modeling year is 2024, and temporal generalization is evaluated only from 2023 to 2024. Multiple market cycles would be needed to evaluate stability under larger interest-rate, policy, and market changes.

Limited model tuning. The study prioritizes controlled experiments over extensive hyperparameter optimization. Systematic cross-validation or Bayesian optimization could improve individual model performance, but would not replace the need for richer explanatory variables.

Unobserved property and neighborhood characteristics. Important attributes such as exact floor, elevator availability, interior condition, renovation quality, noise exposure, distance to transport, school access, neighborhood amenities, and detailed local socio-economic context are absent from the national feature set.

Feature-importance interpretation. Tree feature importances describe how the fitted model uses variables; they do not establish causal effects. In addition, importance can be distributed unevenly across correlated or one-hot encoded predictors.

VII. Conclusion

This study investigated residential property sale-price prediction in France using nationwide open data and a controlled machine-learning protocol. Starting from DVF+ transaction records, a final 2024 national dataset of 708,520 exploitable residential sales was constructed. A naive median baseline, Linear Regression, Random Forest, and Gradient Boosting were compared, after which the incremental contributions of geography, DPE information, and temporal evaluation were examined.

The main research question can be answered in two parts. First, among the tested configurations, the most accurate model according to MAE is Gradient Boosting enriched with department-level geography, with a MAE of approximately EUR 95,345, an RMSE of EUR 222,285, and an R^2 of 0.421. Second, the experiments show that feature information matters at least as much as algorithm choice. Without geography, Gradient Boosting reaches a MAE of approximately EUR 119,235; adding department information reduces this error by roughly 20%.

The four hypotheses lead to differentiated conclusions. H1 is partially supported because Gradient Boosting achieves the best MAE but does not dominate Linear Regression on RMSE and R^2. H2 is strongly supported by the large improvement obtained from department-level geography. H3 is not supported on the matched Parisian sample: the DPE label does not provide an additional predictive gain beyond the selected physical and geographic variables. H4 is moderately supported because training on 2023 and testing on 2024 slightly degrades performance, particularly RMSE.

Finally, the feature-importance analysis shows that dwelling surface and departmental geography dominate the best national model, together accounting for approximately 93.6% of aggregated importance. Future research should therefore prioritize finer spatial context, richer property characteristics, repeated temporal validation, and broader DPE experiments before focusing solely on more complex algorithms.

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